

UGANDA MARTYRS UNIVERSITY NKOZI.

END OF SEMESTER TWO FINAL ASSESSMENT

COURSE UNIT: WEB BASED SYSTEMS PROGRAMMING.

GitHub Repository Link: https://github.com/MugishoRomuald/Studentcourse_Registration-Portal.git

GROUP ONE MEMBERS

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PROJECT REPORT.

1. Project overview.

The Umoja Student Course registration Portal is a web-based application that streamlines course enrollment. Students can browse, search and register for courses while system administrators manage the student course catalogue.

Technologies used: PHP, MySQL, Bootstrap 5, HTML/CSS, JavaScript.

2. Functional requirements.

Requirement	Status	Implementation file
User registration and login	✓	register.php, login.php
User logout and session control	✓	logout.php
Data entry through forms	✓	add_course.php, register.php
Input validation	✓	all form processors
Dynamic data display	✓	courses.php, enrolled.php
Search /filter	✓	bowse_course.php
Editing records	✓	edit.php
Deleting records	✓	delete.php
Role- Based Access	✓	session role checks
Persistent data storage	✓	MySQL database

3. Database design

The database: studentreg_db

Tables with their attributes include:

- i) users(id,name,email,password,role,created_at,updated_at);
- ii) registrations(id,user_id, course_id,status,created_at) with UNIQUE constraints
- iii) courses(id,course_name,course_code,description,created_at,updated_at);

Relationships: users(students)→registrations←courses

4. Security features implemented

Feature	Implementation
Password security	password_hash() bcrypt encryption
SQL injection prevention	Prepared statements for ALL queries
CSRF protection	Tokens on all forms
XSS prevention	htmlspecialchars () on all output
Session fixation	Session_regenerate_id() on login
Role-based access	Session role verification on every page
Admin Secret Key (mugisho)	Required for admin registration

5. User roles and access

Role	Accessible pages	Permissions
Admin	dashboard.php, courses.php, add_course.php, edit.php and delete.php	Create, Read, Update, and Delete courses
Student	studentdashboard.php, browse_courses.php, enroll.php, enrolled.php	Browse, Search, Enroll, View enrolled courses

6. Testing summary

Test case	Result
Admin adds/edits/deletes course	Success
Student enrolls in course	Success
Student searches courses	Filtered results
Duplicate enrollment	Enroll button disabled
Invalid login	Error displayed (Invalid email or password.)
Valid registration	Success
Protected page access without login	Redirected

7. Challenges and solutions

Challenge	Solution
Database connection	We used a different method
Password security	Tried bcrypt hashing
CSRF attacks	Random tokens validated on POST
SQL injection	Prepared statements everywhere

8. Achievements

- Secure authentication. Passwords are hashed using bcrypt hashing, sessions are properly managed, and role-based controls restrict unauthorized access.
- Complete CRUD operations. Administrators can create, read, update, and delete courses with proper validation and confirmation dialogs.
- Student functionality. Students can browse courses, search by name or code, enroll in courses, and view their enrollment history.
- Duplicate prevention. The system prevents students from enrolling in the same course twice through database unique constraints.
- Security implementation. CSRF tokens, prepared statements, XSS prevention, and session fixation protection are implemented throughout.

```
c:\wamp64\bin\mysql\mysql9.1.0\bin\mysql.exe
mysql> users;
+-----+
3 rows in set (0.01 sec)

mysql> DESC courses;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| id | int | NO | PRI | NULL | auto_increment |
| course_name | varchar(100) | NO | | NULL | |
| course_code | varchar(50) | NO | UNI | NULL | |
| description | text | YES | | NULL | |
| created_at | timestamp | YES | | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| updated_at | timestamp | YES | | CURRENT_TIMESTAMP | DEFAULT_GENERATED on update CURRENT_TIMESTAMP |
+-----+
6 rows in set (0.05 sec)

mysql> DESC registrations;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| id | int | NO | PRI | NULL | auto_increment |
| user_id | int | NO | MUL | NULL | |
| course_id | int | NO | MUL | NULL | |
| status | enum('enrolled','dropped') | YES | | enrolled | |
| created_at | timestamp | YES | | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+
5 rows in set (0.01 sec)

mysql> DESC users;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| id | int | NO | PRI | NULL | auto_increment |
| name | varchar(50) | NO | | NULL | |
| email | varchar(150) | NO | UNI | NULL | |
| password | varchar(255) | NO | | NULL | |
| role | enum('admin','student') | YES | | student | |
| created_at | timestamp | YES | | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| updated_at | timestamp | YES | | CURRENT_TIMESTAMP | DEFAULT_GENERATED on update CURRENT_TIMESTAMP |
+-----+
7 rows in set (0.01 sec)

mysql>
```

Manage Courses

+ Add Course

Course Name	Code	Description	Actions
Bachelor of Science in Computer Science	200	This is a 3-year program where students learn and gain skills in various take areas.	Edit Delete
Bachelor of Science in InformationTechnology	300	BSC IT is a 3-year program at our institution	Edit Delete
Introduction to Programming	CS101	Learn fundamentals of programming using Python	Edit Delete

Umoja Student Course Registration Portal



Create Account

Full Name

Email

Password

Register as

Student

Admin

Create Account

[Already have an account? Login](#)

localhost / MySQL / studentcou x Umoja Student Course Registrat x +

← → ↻ http://localhost/studentregistration/enrolled.php ☆ 📁 📄 📌 School ⋮

Umoja Student Course Registration Portal



Hello, Mugisho Muganga

Logout

My Enrolled Courses

[Browse More](#)

Bachelor of Science in Computer Science

200

This is a 3-year program where students learn and gain skills in various take areas.

Enrolled: 06 May 2026

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Windows taskbar: Type here to search, 20:23 08/05/2026, 22°C

localhost / MySQL / studentcou x Umoja Student Course Registrat x +

← → ↻ http://localhost/studentregistration/studentdashboard.php ☆ 📁 📄 📌 School ⋮

Umoja Student Course Registration Portal



Hello, Mugisho Muganga

Logout

Welcome, Mugisho Muganga!

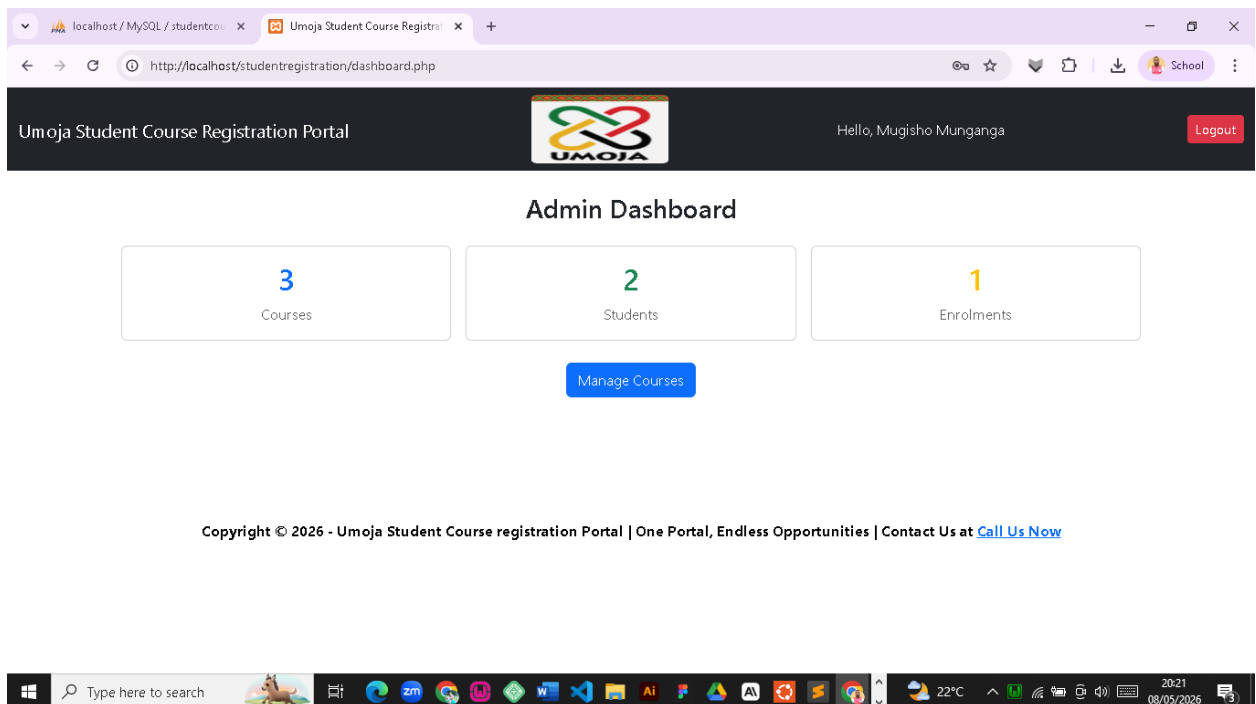
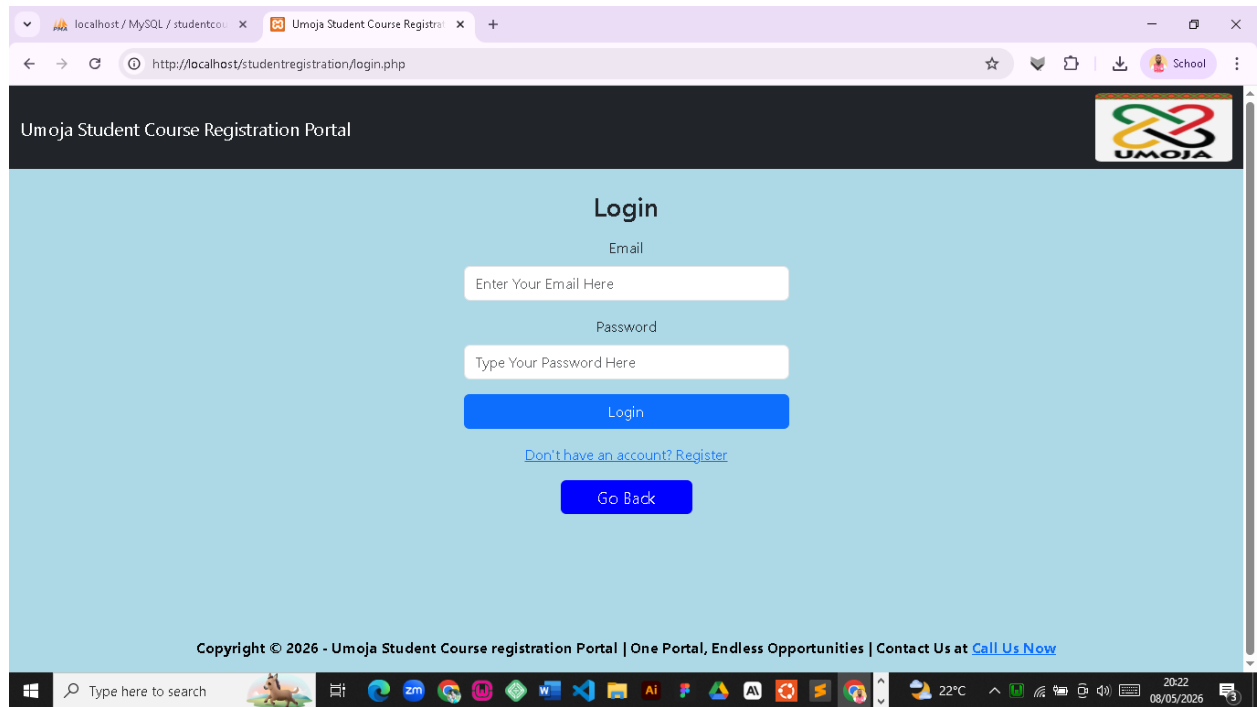
1

Courses enrolled

[Browse Courses](#)[My Enrolled Courses](#)

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Windows taskbar: Type here to search, 20:23 08/05/2026, 22°C



9. Conclusion.

We conclude by thanking our instructor in this course unit for the continuous guidance in our academic endeavors. This task was not only marked or contributed to our end of semester II exam but also introduced to us on how we can use the knowledge obtained from class to solve real-world problems in our communities. We had the opportunity to combine the classroom knowledge together with that we have been gaining through watching

endless tutorials. Thank once again sir for this opportunity. We realized that we really learnt a lot.